

MASTER THEOREM FOR DIVIDING FUNCTIONS

①

- The master theorem is used in calculating the time complexity of recurrence relations (divide and conquer algorithms) in a simple and quick way.
- The master method is a formula for solving recurrence relations of the form:

$$T(n) = a T(n/b) + f(n)$$

where,

n = size of input

a = number of subproblems in the recursion

n/b = size of each subproblem. All subproblems are assumed to have the same size.

$f(n)$ = cost of the work done outside the recursive call, which includes the cost of dividing the problem & cost of merging the solutions.

→ Here, $a \geq 1$ and $b > 1$ are constants, and

$f(n)$ is an asymptotically positive function

$$f(n) = O(n^k \log^p n)$$

②

→ first, we have to find, two values:

① $\log_b^a$

② k

→ There are three cases:

Case 1:

if $\log_b^a > k$ then $O(n^{\log_b^a})$

Case 2:

if $\log_b^a = k$

(i) if $P > -1$ then $O(n^k \log^{P+1} n)$

(ii) if $P = -1$ then $O(n^k \log \log n)$

(iii) if $P < -1$ then $O(n^k)$

Case 3:

if $\log_b^a < k$

(i) if $P \geq 0$ then $O(n^k \log^P n)$

(ii) if $P < 0$ then $O(n^k)$

Examples (Case 1) : Solve the following recurrence relation using master's theorem.

① $T(n) = 2T(n/2) + 1$

Sol :

Here, $T(n) = aT(n/b) + f(n)$

(i) $a=2, b=2$

$f(n) = \Theta(1)$

$= \Theta(n^0 \log^0 n)$

$\therefore f(n) = O(n^k \log^p n)$

Here, $k=0, p=0$

(ii) $\log_b^a = \log_2^2$

$= 1$

$\left. \begin{array}{l} \log_b^a > k \\ 1 > 0 \end{array} \right\}$

It satisfies case 1.

Hence,

$T(n) = O(n^1)$

$$(2) \quad T(n) = 4T(n/2) + n$$

Sol:

$$\begin{aligned} T(n) &= aT(n/b) + f(n) \\ f(n) &= O(n^K \log^P n) \end{aligned}$$

Here,

$$(i) \quad a = 4, \quad b = 2$$

$$f(n) = n^1 \log^0 n$$

$$K = 1, \quad P = 0$$

$$(ii) \quad \log_b a = \log_2 4 = 2$$

$2 > 1$ Hence, case 1 satisfied

Hence,

$$T(n) = O(n^2)$$

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③ $T(n) = 8T(n/2) + n$

Sol:

(i) $a = 8, b = 2$

$f(n) = n^1 \log^0 n$

$K=1, P=0$

(ii) $\log_b a = \log_2 8 = 3$

$\boxed{3 > 1}$. Hence case 1 is satisfied.

$T(n) = \Theta(n^3)$

④ $T(n) = 8T(n/2) + n^2$

Sol:

(i) $a = 8, b = 2$

$f(n) = n^2 \log^0 n$

$K=2, P=0$

(ii) $\log_b a = \log_2 8 = 3$

$\boxed{3 > 2}$. Hence case 1 is satisfied.

$T(n) = \Theta(n^3)$

Examples: case 2

① $T(n) = 2T(n/2) + n$

Sol:

(i) $a=2, b=2$

$f(n) = n^1 \log^0 n$

$K=1, P=0$

(ii) $\log_b a = \log_2 2 = 1$ K=1

$\log_b a = K$

$1=1$

Hence, case 2 is satisfied.

(iii) $P > -1$

$0 > -1$

Hence, $\Theta(n^K \log^{P+1} n)$

i.e. $\Theta(n^1 \log^1 n)$

$T(n) = \Theta(n \log n)$

$$(2) T(n) = 2T(n/2) + n \log n$$

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Sol:

$$(i) a = 2, b = 2$$

$$K = 1, P = 1$$

$$(ii) \log_b^a = \log_2^2 = 1$$

$$\left. \begin{array}{l} \log_b^a = K \\ 1 = 1 \end{array} \right\} \text{Hence, case 2 is satisfied:}$$

$$(iii) P > -1$$

$$1 > -1$$

$$\text{Hence } T(n) = \Theta(n \log^2 n)$$

$$(3) T(n) = 4T(n/2) + n^2$$

Sol:

$$(i) a = 4, b = 2, K = 2, P = 0$$

$$(ii) \log_b^a = \log_2^4 = 2$$

$$\log_b^a = K$$

$$2 = 2$$

$$(iii) P > -1$$

$$0 > -1$$

$$\text{Hence } T(n) = \Theta(n^2 \log n)$$

$$T(n) = 4T(n/2) + n^2 \log n$$

$$\Theta(n^2 \log^2 n)$$

$$(4) T(n) = 2T(n/2) + \frac{n}{\log n}$$

Sol:

$$(i) a=2, b=2, K=1, P=-1 \quad \underline{(n' \log^{-1} n)}$$

$$(ii) \log_b a = \log_2^2 = 1$$

$$\left. \begin{array}{l} \log_b a = K \\ 1 = 1 \end{array} \right\} \text{Hence, case 2 is satisfied}$$

$$(iii) P = -1$$

$$-1 = -1$$

$$\text{Hence, } T(n) = \Theta(n' \log \log n)$$

$$(5) T(n) = 2T(n/2) + \frac{n}{\log^2 n}$$

Sol:

$$(i) a=2, b=2, K=1, P=-2$$

$$(ii) \log_b a = \log_2^2 = 1$$

$$\log_b^a = k \left. \begin{array}{l} \\ 1 = 1 \end{array} \right\}$$

Hence, case 2 is satisfied.

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$$(iii) P < -1$$

$$-2 < -1$$

Hence, $\Theta(n)$

Examples: case 3

$$\textcircled{1} T(n) = T(n/2) + n^2$$

Sol:

$$(i) a = 1, b = 2, k = 2, P = 0$$

$$(ii) \log_b^a = \log_2^1 = 0$$

$$\textcircled{+} \log_b^a < k \left. \begin{array}{l} \\ 0 < 2 \end{array} \right\}$$

case 3 is satisfied

$$(iii) P \geq 0$$

$$\Theta(n^2 \log^0 n)$$

$$\therefore \Theta(n^2)$$

$$(2) T(n) = 2T(n/2) + n^2 \log^2 n$$

Sol:

$$(i) a = 2, b = 2, K = 2, P = 2$$

$$(ii) \log_b^a = \log_2^2 = 1$$

$$\left. \begin{array}{l} \log_b^a \leq K \\ 1 < 2 \end{array} \right\} \text{Case 3 is satisfied}$$

$$(iii) P \geq 0$$

$$2 \geq 0$$

$$\text{Hence, } \Theta(n^K \log^P n)$$

$$T(n) = \Theta(n^2 \log^2 n)$$